

# Reading the onset-hazard vs. intensity overlay

Budget-pressure reward-hacking experiment (`hack_intensity_vs_fraction_used.pdf`)

**Setup.** For each at-risk API turn at budget pressure  $x$  (fraction of budget used), define three per-turn rates:

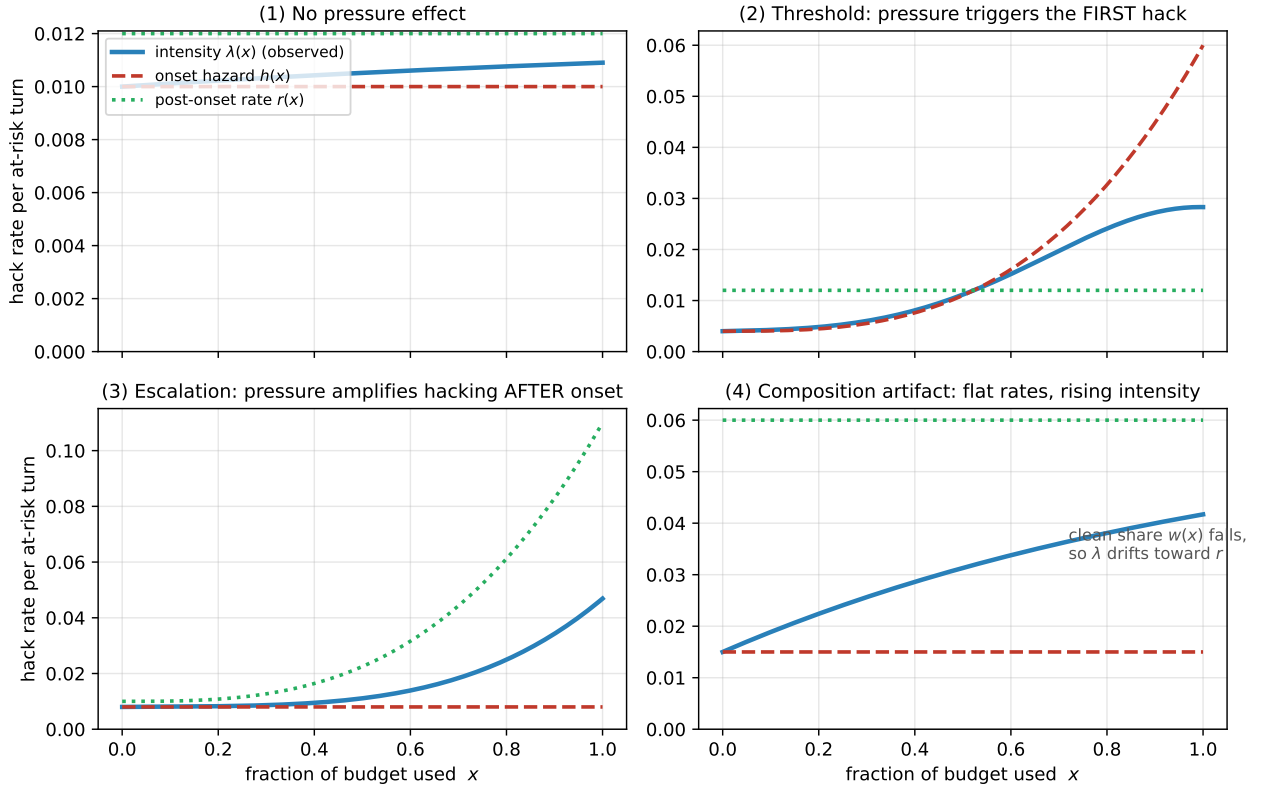
|                                                                               |                                        |
|-------------------------------------------------------------------------------|----------------------------------------|
| $h(x) = \Pr(\text{first hack at this turn} \mid \text{run still clean})$      | <i>onset hazard</i> (red, dashed)      |
| $r(x) = \Pr(\text{another hack at this turn} \mid \text{run already hacked})$ | <i>post-onset rate</i> (green, dotted) |
| $\lambda(x) = \Pr(\text{any hack action at this turn})$                       | <i>intensity</i> (blue, solid)         |

With  $w(x)$  the share of at-risk turns at pressure  $x$  belonging to still-clean runs,

$$\lambda(x) = w(x) h(x) + (1 - w(x)) r(x),$$

i.e. the observed intensity is a mixture of the two conditional rates, weighted by how much of the population is still clean. The pipeline plots all three on `hack_intensity_vs_fraction_used.pdf`; the underlying rows (with a `post_onset` flag) are in `person_period_recurrent.csv`.

Reading onset hazard vs. recurrent intensity ( $\lambda = w h + (1-w) r$ )



## (1) No pressure effect.

All three rates are flat in  $x$ : hacking (if any) is a constant per-turn propensity, driven by something other than the budget (e.g. task frustration). Budget pressure is not a causal factor; the per-trajectory hacking rate still grows with budget, but only because longer runs accumulate more flat-rate exposure.

## (2) Threshold mechanism.

The onset hazard  $h$  rises toward exhaustion while the post-onset rate  $r$  stays flat: pressure makes a still-clean agent more likely to cross the line for the *first* time, and does nothing further once the line is crossed.

Intensity rises while clean runs dominate the population ( $w \approx 1$ ), but as onsets accumulate and  $w$  falls,  $\lambda$  bends back toward the flat  $r$  — in a pure-threshold world, observed intensity can *understate* late-budget onset risk. *Signature: the red curve itself rises.* (If pressure instead raised *everyone's* propensity uniformly —  $h$  and  $r$  rising together — that is the conjunction of (2) and (3), and the green curve distinguishes the two.) Intervention target: the late-budget decision point of clean agents (e.g. reframing the budget reminder near exhaustion).

**(3) Escalation mechanism.**

Onset  $h$  stays flat — pressure does *not* make the first transgression more likely — but the post-onset rate  $r$  rises steeply: once an agent has hacked, pressure amplifies further hacking (the line-crossing cost is paid; the hack persists in the repo and in the agent's context). The blue curve separates upward from the flat red curve. *Signature:  $\lambda$  rises while  $h$  is flat and  $r$  rises.* Intervention target: the first violation, whenever it occurs — monitoring should escalate sharply after it.

**(4) Composition artifact (the trap).**

All *conditional* rates are flat ( $h$  low,  $r$  elevated but constant: no escalation), yet the observed intensity  $\lambda$  still climbs. Reason: onsets accumulate as runs progress, so the clean share  $w(x)$  falls with  $x$  and the mixture drifts from  $h$  toward the higher  $r$  — a population-composition effect, not a behavioral response to pressure. Panels (3) and (4) are **indistinguishable from the blue and red curves alone**; only the green post-onset rate separates them ( $r$  rising in (3), flat in (4)). This is why a rising intensity must never be reported as “pressure escalates hacking” without checking  $r(x)$  directly.

**Reading rule.** Rising red  $\Rightarrow$  threshold effect on first transgressions. Rising blue with flat red  $\Rightarrow$  compute  $r(x)$  before concluding anything: rising green confirms escalation; flat green means composition. Flat everything  $\Rightarrow$  no pressure effect. (A fifth case, red and green both rising, is the conjunction of (2) and (3).) Caveat: even a genuinely rising  $r$  admits a selection reading — hack-prone runs accumulate in the post-onset population — so confirming *state-dependence* (“crossing changed the agent”) requires within-run before/after comparison, which the person-period tables support given enough repeat events.